Don't blame the tech

I'm sure all of you have been following the tragic but mysterious circumstances of the disappearance of Malaysian Airlines MH370. Every one I know, including many of us here at Digit have been left flabbergasted by the sheer unexpectedness of this incident – how do you lose an airplane? This is 2014, right?

How does a commercial airplane, which has already been proven to be a potent missile by Al-Qaida nearly 13 years ago, drop off the sensors of not just regular air traffic control, but also military scanners? It did, after all, divert from its flight path, not respond to communications and had ACARS (Aircraft Communications Addressing and Reporting System) and its transponder turned off! Am I the only one who pictures frantic war rooms and fighter jets being scrambled?

The latest news (as I write this) is that satellite data puts the last known location of the plane somewhere in the southern Indian Ocean. Given how hard it is to salvage anything from the middle of an ocean, we have to accept that we may never have satisfactory answers for what really happened. Technology has failed, spectacularly, or has it?

Take the simple example of the ACARS system, which is basically a text-based system that tells ground control the location, speed, altitude and direction of a flight. Why should such an important system have a switch or circuit-breaker that a pilot can throw on or off? Aren't we giving the pilot of a plane way too much power? Controls, yes, I wouldn't want to be in a plane flown by wire from the ground. Give me a human pilot any day, but allowing them to turn off critical tracking systems? Perhaps there's some logic for it, but I think it's time to re-look at established logic.

The transponder is another example of technology that needs improvement, because it relies solely on ground-based devices asking it a question (in machine language) to which it responds. The range is limited however – about 100 kilometres. As long as you're flying over populated land masses, the transponder works well, but over large bodies of water it is pointless. Again, pilots have a simple switch to turn the transponder off, even in mid-air, why?

While the black box is mostly foolproof, it only records the status of equipment detecting whether there's been a mechanical or system failure – but in this case it might read something like, "All systems go. Cause of crash: ran out of fuel". What answers about MH370 does that give us? Nothing.

That's where we turn to the cockpit voice recorder for answers, which, however, might also be useless, because the voice recorder only records the last two hours of conversation between the pilots, and not the whole journey! Why? Because we've convinced ourselves that crashes happen immediately after a problem, and thus two hours suffice.

I'm aware that such systems have a myriad level of complications when being designed, but still, surely even a few GB of storage can record an entire flight's conversation! I can record 100 hours of HD video on my cheap DVR, but planes that cost tens of millions of dollars can record only two hours of audio?

I guess there are two major reasons this story has struck a chord with me:

First, I belong to a family who has grieved a relative "lost at sea", and thus I can relate to the pain of the families of those on board MH370. Back then, I was asking how 21st century, multi-million dollar merchant navy ships still don't have basic technology such as CCTV cameras observing the deck to make sure that just in case someone fell overboard, the company would be able to provide proper evidence of what transpired. The technology exists, is used all around us, but sometimes isn't implemented where it's needed most.

Second, because at about the time MH370 went missing, I booked a holiday to Australia for my family on Malaysian Airlines, flying Boeing 777-200s...

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